

# 11. Respiration-Osmoreguation

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## Pre-class materials

**i** Read ahead

**Before class, you can prepare by reading the following materials:**

1. Wednesday we will finish respiration and Friday begin osmoregulation.
2. The Symmorphosis paper is a very interesting and important paper. It will be useful for design3 respiration.
3. In Lab this week Tuesday you will do Toad Muscle Independent labs, Wednesday you will measure your own brain waves using EEG. Have fun! [\[manual\]](#)!

## Announcements/Reminders

- Due Friday at the start of class– Homework 5 Q1,2,5 [\[schedule\]](#)
- Due Monday 11/25 at midnight - Design 3 draft
- Please do your peer review for design 3 by Wednesday and send to colleagues and myself
- Do discussion TEAMMATES eval, released each Friday, due by Monday.

## Week 13 Discussion Groups

| Group | Partner 1 | Partner 2 | Partner 3 |
|-------|-----------|-----------|-----------|
| 1     | Manuel    | Trey      | Reagan    |
| 2     | Kyra      | Carla     | Emma      |
| 3     | Peigi     | Niki      | Darwin    |
| 4     | Alice     | Dallas    | Paris     |

## Wednesday -

### Aquatic Respiration, Fish Gills

- **Reading assignment:** Withers Aquatic respiration: skim 565-72, read 573-4, 585-99, supplement 12-2. ALTERNATIVELY, read HWA chapter 22 + HWA 586-587 (counter-curr) + HWA 590-594 (fish).
- [discussion pg 3, Q4-5] [slide deck] [slide deck2]
- **Topics:**
  - Partial pressures of gasses and pO<sub>2</sub> at sea level - the starting point.
    - \* Influences of relative humidity
    - \* Altitude (total pressure)
    - \* Henry's Law - dissolved O<sub>2</sub> in water
  - Flow patterns: Countercurrent vs. Concurrent
  - Fish Gills
    - \* Ventilatory pumps in fish (breathing patterns)
  - Root Effect

Respiratory Pumps

<https://youtu.be/2GPfu8ebZac>

### Finish aerial respiration questions 2, 3, 4

- **Reading assignment:** Withers pp. 609-631 OR HWA chapter 23 + Withers 626-632, skip invertebrates
- [discussion pg 4:2,3,5] [slide deck2]
- **Aerial Respiration Topics:**
  - Air flow patterns of vertebrates

Respiration the Movie

<https://youtu.be/clyu9h810n4>

## Friday - Symmorphosis

- **Reading assignment:** Symmorphosis: Weibel, Taylor, and Hoppeler (1991) The concept of symmorphosis: A testable hypothesis of structure-function relationship. PNAS 88:10357-61 [Respiration Papers folder](#)
- In class discussion

## Begin Osmoregulation

- **Reading assignment:** Withers skim beginning Ch. 16 for definitions in list below, also read Water & Ion Budgets pp. 788-790, Vertebrates pp. 798-803, Terrestrial Environments pp. 806-812, Terr. Verts. 822-827. Withers is actually very readable on this topic, and some of it will be familiar to you already:).
- [discussion] [slide deck] [slide deck 2]
- definitions:
  - solutes,
  - osmosis,
  - osmolarity/osmolality,
  - ionoregulation,
  - osmoregulation/osmoconforming,
  - compatible solutes,
  - perturbing solutes,
  - counteracting solutes,
  - osmotic permeability (& formulae)
- The components of **Water Budgets** (input/output) as well as **Ion Budgets**
- The water and ion challenges of **freshwater**, **marine**, and **terrestrial** environments

Osmoregulation Podcast

<https://youtu.be/1vQxLFoPCmM>

### **i** Coming up Next Week

- Tuesday Lab: EEG lab.
- Wednesday class: work on your own with your partner on Design 3 (you may work in the classroom, in the library, etc.).
- Wednesday Lab: No lab
- Submit your lab reports to your TA by email at the regular time.
- Friday: Holiday!
- Happy Thanksgiving!